Arghamitra Talukder

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Education

2023 - present	PhD in Computer Science at **Columbia University** *NSF CSGrad4US Fellow*	(GPA: 3.98/4.0)
2018 - 2021	BS in Electrical Engineering at **Texas A&M University** *Engineering Honors, Magna Cum Laude*	(GPA: 3.78/4.0)

Projects

Predicting Exon Inclusion Levels via Contrastive Sequence Models

Developed a deep learning model to analyze gene regulation with a contrastive learning framework. This model identifies conserved functional elements by mapping evolutionarily related DNA sequences into a shared latent space, providing novel insights into gene expression.

Bayesian hierarchical isoform quantification with hybrid sequencing

Developed JOLI, a novel hierarchical model that enhances RNA transcript quantification by jointly integrating short-read (SR) and long-read (LR) sequencing data. The model utilizes a multi-sample learning framework to improve estimation accuracy and reproducibility, particularly for low-abundance isoforms.

Modeling cell plasticity with integrated single-cell data

Designed a statistical model to quantify cellular plasticity by integrating single-cell lineage and gene expression data. This approach utilizes a continuous annotation distribution methodology to statistically detect and score plastic cell states.

Publications

Talukder, Arghamitra, Shree Thavarekere, et al. (2025). "Multi-sample, multi-platform isoform quantification using empirical Bayes". In: *RECOMB-Seq (Short-talk)*. URL: https://www.biorxiv.org/content/10.1101/2025.02.08.637184v2.abstract.

Talukder, Arghamitra and Itsik Pe'er (Sept. 2024). "Computational Method to Detect Cancerous Plasticity". In: *IICD Workshop on Single-Cell Data Integration & OT, CRA-WP Grad Cohort for IDEALS (Short-talk)*.

Talukder, Arghamitra, Rujie Yin, et al. (2022). "Does inter-protein contact prediction benefit from multi-modal data and auxiliary tasks?" In: *MLSB, NeurIPS*. URL: https://www.mlsb.io/papers_2022/Does_Inter_Protein_Contact_Prediction_Benefit_from_Multi_Modal_Data_and_Auxiliary_Tasks.pdf.

Ibrahim, Bassem, **Talukder, Arghamitra**, and Roozbeh Jafari (2020). "Multi-source multi-frequency bio-impedance measurement method for localized pulse wave monitoring". In: *EMBC, IEEE*. URL: https://ieeexplore.ieee.org/abstract/document/9176495.

Fellowships

NIH T15 Training Grant — Sep 2025 - Aug 2026
CISE CSGrad4US Fellowship, CRA, NSF — Sep 2023 - Aug 2027
NeurIPS Travel Award — Dec 2022

Work Experience

Product & Test Engineer at Texas Instruments, Dallas, TX — June 2021 - Aug 2023

- Responsible for TI customed Ultra-High Voltage test technology with 70% cost efficiency.
- Created a product specific test program converting from legacy tester to the latest technology reducing 40% of the test time.

Test & Validation Intern at Texas Instruments, Dallas, TX — June 2020 - Aug 2020

- Enabled traceability with the accuracy of 99.9% by device ID recovery obtained using OCR technology.
- Automation of data acquisition and validation by database access and scripting using SQL and PERL.

Teaching Assistant

Machine Learning for Functional Genomics, Columbia University — Sep 2025 - Dec 2025
Computational Genomics, Columbia University — Jan 2025 - May 2025
Experimental Physics Lab II-Mechanics, Texas A&M University — Jan 2019 - Dec 2019
Foundation of Engineering in Python and LabVIEW, Texas A&M University — Sep 2018 - Dec 2018

Volunteering

Conference volunteer at MLCB — Sep 2025
Columbia Pre-submission Application Review (PAR) — Nov 2024
Critical Thinking in STEM career panel 2024 — June 2024